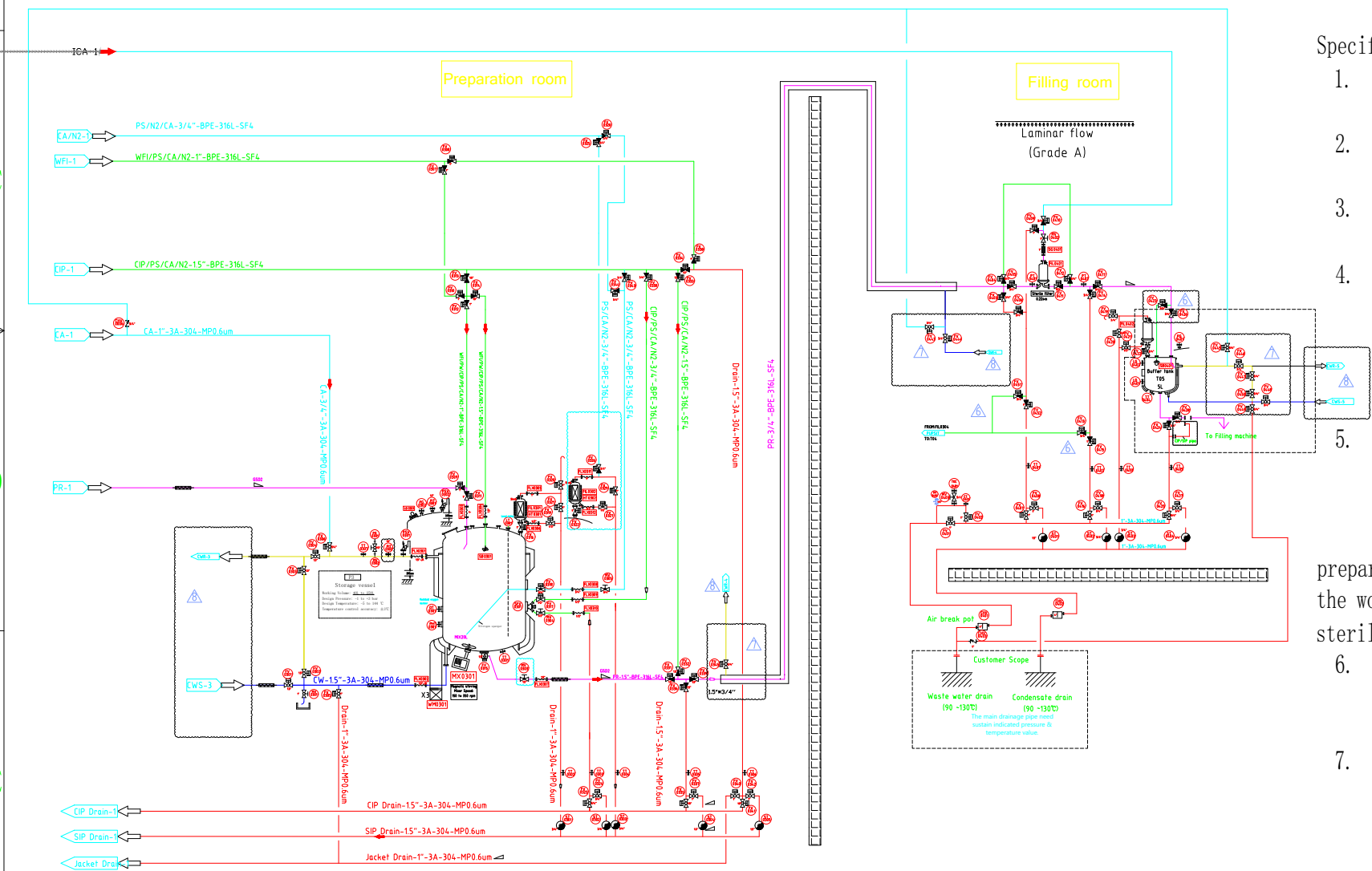
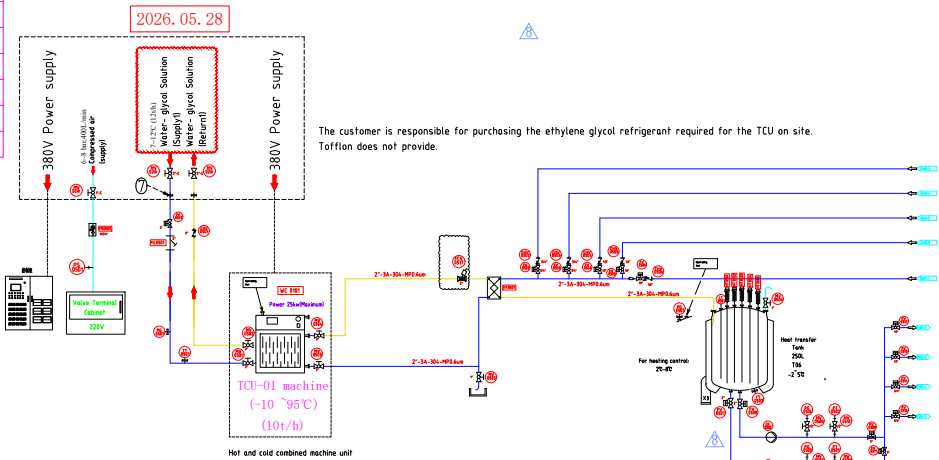


1			2			3			4					
TAG	SIGN	NAME	TAG	SIGN	NAME	TAG	SIGN	NAME	TAG	SIGN	NAME	TAG	SIGN	NAME
PS		Pneumatic ball valve	MB		Manual ball valve	RD		Rupture disk	APR		Air pressure regulator	CP		Centrifugal pump
PS		Pneumatic diaphragm valve	MD		Manual diaphragm valve	VL		View window with light	ATD		Air treatment device	SP		Self-priming pump
ST		Steam trap	CV		Check valve	VM		View window	LC		Load cell	DP		Diaphragm pump
PS		Pneumatic bottom valve	MB		Manual bottom valve	PM		Pipe view mirror	SSD		Syringes sampling device	PP		Peristaltic pump
PS		Pneumatic sampling valve (single port)	MS		Manual sampling valve (single port)	VS		Vessel sight glass	EH		Electric heater	TLP		Twin lobe pump
PS		Pneumatic sampling valve (double port)	MS		Manual sampling valve (double port)	SH		Sanitary hose	EVH		Electric heater for vent filter	DC		Drain cooler
PS		Pneumatic butterfly valve	MB		Manual butterfly valve	YS		Y-type strainer	AHC		Air filter housing	PHE		Plate heat exchanger
PS		Pneumatic diaphragm proportional control valve	SP		Steam pressure regulator valve	SSB		Static spray ball	AFC		Air filter core	DSTHE		Double tube sheet heat exchanger
PS		Pneumatic angle seat proportional control valve	SV		Safety valve	DSB		Dynamic spray ball	WRVP		Water ring vacuum pump	V		Vacuum pump
AB		Air breaker	CL		Clamping-type sampling valve	SM		Static mixer	F		Fan			



Specification:

- The prepared solution is easy to dissolve and viscosity is low (water-like). Equip low force magnetic agitator for tanks for mixing and dissolving solution.
- Equip a WFI heat exchanger for WFI dosing for instantaneous cooling (From 85°C to 25°C) to save time. Tank jacket is used to control temperature in vessel.
- The preparation vessel solution transfer is done by using compressed air/ nitrogen and the holding vessel solution transfer is done by using compressed air/ nitrogen.
- The sterilization filter can do integrity test in place(before and after use). Solution filters (in preparation system) will conduct integrity test automatically in place. Waste water of integrity test will drain to T05 10L air tank; The police filter (in the isolator) can conduct integrity test automatically in place, Waste water of integrity test will drain to T04 filling buffet tank(The filling buffer tank will be purged after the test).
- When preparation tank is doing CIP/SIP and drying by compressed air, the holding tank can do filling by compressed air/ nitrogen; When holding tank is doing CIP/SIP and drying by compressed air, the preparation tank can do preparation and do nitrogen bubbling. The P1 and P2 preparation vessels can be configured such that when the P1 vessel is in the preparation stage, the P2 vessel is in the cleaning and sterilization process.(The system cannot achieve the working conditions where the preparation vessel is being cleaned and the storage tank is undergoing sterilization)
- This preparation system consists of functions including dosing WFI, solution mixing, temperature control by jacket, nitrogen bubbling, solution sampling, solution transfer, solution filtration and integrity test(Online integrity test of liquid filter, offline integrity test of air filter).
- CIP step: PW rinsing, Detergent circulation cleaning,WFI rinsing, conductivity judgment for cleaning effect.

NO.	STANDARD NO. / DRAWING NO.	DESCRIPTION	MATERIAL	QTY	UNIT	TOTAL	REMARKS
					MASS (kg)		
Tofflon			THIS DRAWING IS COPYRIGHT. IT MUST NOT BE COPIED, REPRODUCED OR OTHERWISE MADE USE OF WITHOUT THE TOFFLON'S WRITTEN PERMISSION.				
TITLE: Solution preparation system APLS-250L				SHEET: 02 OF: 02		SIZE: A3	
				DWG: 2026-012Pc-P&ID		REV: 02	
CUSTOMER: Promomed Biokhimik				PROJECT NUMBER: 2026-012Pc			
DESIGNED: Alan Hu	DATE: 2026.06.15	MASS: NA	TOLERANCED UNLESS OTHERWISE NOTED				
CHECKED: Dong Wang	DATE: 2026.06.15	SCALE:					
APPROVED: Ziyi Feng	DATE: 2026.06.15	MATERIAL:					
				MACHINING	WELDING	SHEET METAL	
				GB/T 1804-m GB/T 1184-K	GB/T 19804-B GB/T 19804-F	GB/T 15055-m	